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CASE REPORT



Basic principles for the development of an AI-based tool for assistive technology decision making

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ABSTRACT

Introduction: The impact of assistive technology use on the lives of people with disabilities has long been demonstrated in the literature. Despite the need for assistive technologies, and a wealth of innovative, affordable, and accessible products, a low rate of assistive technology uptake is globally maintained. One of the reasons for this gap is related to data and knowledge formation and management. Low access to information and a lack of assessment services is evident. Fragmentation of data, inconsistency in assessment methodology and heterogeneity in the competence of assistive technology professionals, has led to a growing interest in the opportunities that data sciences, including AI, hold for the future of the assistive technology sector, as a supportive and constructive mechanism in any decision-making process.

Objectives: In this short paper, we seek to describe some of the principles that such an AI-based recommendation system should be built upon, using the Atvisor platform as a case study. Atvisor.ai is an AI-based digital platform that supports assistive technology assessments and the decision-making process.

Recommendations: Our recommendations represent the aggregated insights from two pilots held in Israel, testing the platform in multiple environments and with different stakeholders. These recommendations include ensuring the continuum of care and providing a full user journey, incorporating shared decision making and self-assessment features, providing data personalisation and a holistic approach, building a market network infrastructure and designing the tool within a wider service delivery model design. Assessment and decision-making processes, crucial to optimal uptake, can be leveraged by technology to become more accessible and personalised.

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market network

► IMPLICATIONS FOR REHABILITATION

- Provides principles for the development of an AI-based recommendation system for assistive technology decision making.
- Promotes the use of artificial intelligence to support users and professionals in the assistive technology decision making process.
- Personalization of data regarding assistive technology, according to functional, holistic and client centered profiles of users, ensures optimal match and better use of assistive technology.
- Self-assessment and professional assessment components are important for enabling multiple access points to the assistive technology decision making process, based on the preferences and needs of users.

The potential outcomes of assistive technology use and the impact it can have on the lives of people with disabilities have long been discussed and demonstrated in the literature (for recent examples see Refs. [1–6]). The COVID-19 pandemic and the introduction of social distancing and quarantine, led to at-risk populations, including those with disabilities, to be tied to their homes and has brought an added urgency to the planning and provision of assistive technologies. Direct and personal services must maintain a safe and independent distance, despite a growing need to find and distribute assistive technologies that function as effective support.

In a 2019 global consultation, Chapal Khasnabis, the head of the Global Cooperation for Assistive Technology (GATE) of the WHO, stated that assistive technology is currently characterised by “a need, but there is no demand” [7]. Indeed, despite an

evident need for assistive technologies from the user’s perspective, and a wealth of innovative, affordable and accessible products in an open market of possibilities and mainstream solutions, this has not translated into common rules of provision, supply and demand [8]. As a result, the low rate (5–15%) of assistive technology uptake is globally maintained [9].

One of the reasons for this gap is related to data and knowledge formation and management. Low access to independent information and a lack of assessment services for assistive technology is evident. Assistive technology, unlike many commodities or products, requires judgement in identifying appropriate solutions and further customisation and support in training and usage. This is dependent on the goals and needs of the person seeking a solution and the complexity of the solution itself. Comprehensive work in the field of matching person to

technology [10–12] has shown that many elements must be considered in choosing an assistive technology solution. Assistive technology professionals, often responsible for facilitating such a decision-making process together with their clients, may also struggle with the complexities described (for example Arthanat et al. [13]).

Fragmentation of knowledge and information, inconsistency in assessment methodology and heterogeneity in the competence of assistive technology professionals, has led to a growing interest in the opportunities that data sciences, including AI, hold for the future of the assistive technology sector, as a supportive and constructive mechanism in any decision-making process [14].

In this short paper, we seek to specify and describe some of the principles that such an AI-based recommendation system should be built upon, using the Atvisor platform as a case study and an example.

Atvisor.ai is an AI-based digital platform that supports assistive technology assessments and the decision-making process. Used by assistive technology professionals and users, it serves as a CDSS (clinical decision support system), to help in structuring the decision-making process, providing relevant ideas for assistive technology solutions and enables purchasing of the selected product online. Usage begins with a simple search in the assistive technology catalogue and may proceed to a more personalised profile-based search (Figure 1).

The end-user, supported by the assessor, defines his/her unique profile in keeping with the ICF as a bio-psycho-social analysis. Following the creation of a personal profile, the user can then choose the personal goals that are central to him/her and for which activities assistive technology is required. For each goal, the platform creates a personal report, containing recommendations across different assistive technology categories and subcategories, that are relevant to the profile and the activity mentioned.

The end-user can review the report, together with the assessor and independently, consult with family and friends and select the product he/she would like to buy or have provided. Online

purchasing can be integrated within the platform, alongside information about local shops, suppliers and lending programs. Relevant products can be shared, reviewed or marked as favourites for continuous use, while information from searches, recommendations, purchases and online correspondence is stored both in the personal archive of the end-user and that of the assessor. This enables follow up and long-term support, as well as re-initiation of further personalised assistive technology searches for any additional goals.

On the technological level, Atvisor had developed a new way of analysing assistive technology products, based on client needs. The possibility to manage an online matching between the user and the technology, is enabled through a personal matching component that contains

- an infrastructure of rules sets emulating the thinking process of a senior professional in the assistive technology field, supporting it and expanding it
- a matching algorithm that “connect” between the profile of the person and the right assistive technology solutions
- a functional assessment tool enabling the definition of a personal profile

The following recommendations represent the aggregated insights from two pilots held in Israel, testing the platform in multiple environments and with different stakeholders¹:

1. Ensure the continuum of care and provide a full user journey In the assistive technology process, there are many “entry and exit” points, where the assistive technology user can deal with questions and concerns, which if not addressed, may lead to product abandonment. Following the iterative and repeated cycles of product testing and user feedback, by assistive technology professionals and users engaged in, the following stages have been identified:
 - **Discovery** (general exploration of possibilities, sometimes with no specific goals declared, and to “open up” the possibility of using assistive technology),

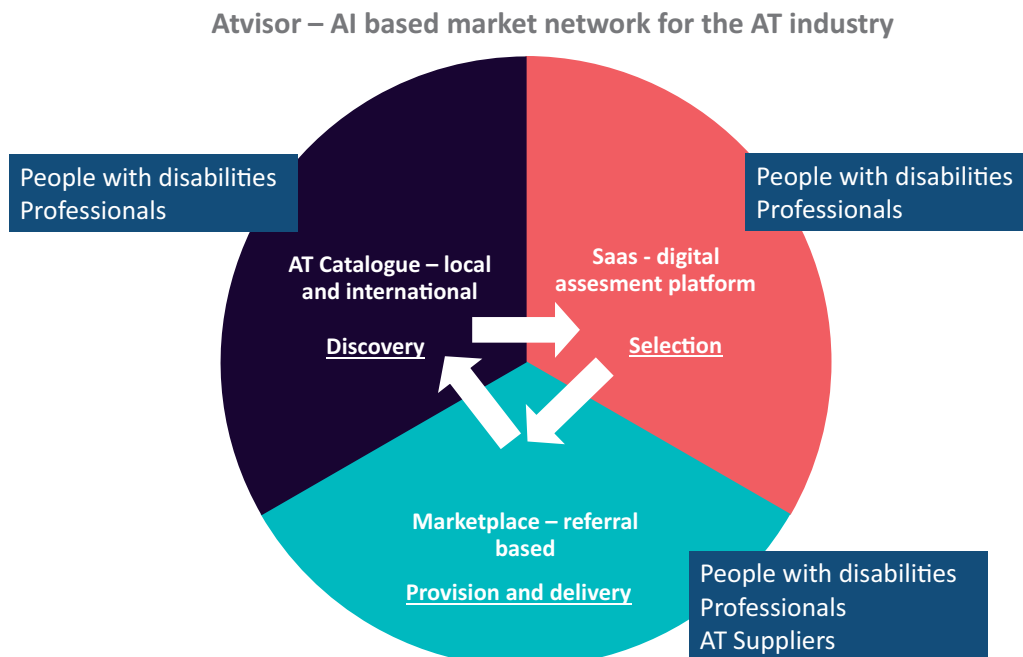


Figure 1. A suggested Market Network model for the Atvisor platform including three platform components – database (catalogue), digital assessment platform (SaaS) and a marketplace, as well as presenting three stakeholders communities using the platform – assistive technology users, professionals and suppliers.

- **Identification** (focusing or “zooming in” on assistive technology categories and subcategories that might be of relevance to the current search),
- **Selection** (interest that is shown regarding specific or types of assistive technology products. This may entail loaning and trial possibilities before reaching the final decision),
- **Provision** of products from an assistive technology seller,
- **Delivery** of the purchased product and beginning of use.

With the understanding of the value that systems and processes give to users if support is present throughout [15], three distinct yet interconnected modules in the Atvisor platform have been developed – the *discovery module*, the *assessment module* and the *provision and follow up support module*. By moving forward from module to module across the platform, the assistive technology user and professionals can be digitally supported throughout the identification of needs, the selection of products, the provision of the desired solution and its optimal use.

As the platform offers many potential solutions to each goal or challenge presented, the professional can maintain involvement until completion, without creating an unintentionally biased or limited recommendation for a specific product.

2. Incorporation of shared decision making and self-assessment features

Current effective practice assumes that the assistive technology user will be involved in the decision-making process and that a user-centred service, will eventually lead to better technology uptake by the person [16,17]. Nevertheless, as learned from pilot results, assistive technology professionals and users might conceive shared decision-making processes quite differently. This discrepancy is significant and influential. The understanding of its mechanisms is recommended as a matter for further and future research. It appears that the use of a digital tool that accompanies stakeholders as a step by step experience can be an effective way to construe and unify such variance. Based on that understanding, the next step is a move towards self-assessment, for situations and products where professional involvement is less critical. In Atvisor the Self-assessment feature provides a better opportunity for self-determination and expression of needs and goals by assistive technology users. It demonstrates in practice, the move towards participative and affirmative models of disability while easing pressure on professionals’ case-load and releasing resources for cases where their involvement most essential.

3. Personalisation

An optimal match between the needs and characteristics of those with a disability and the most appropriate assistive technology requires a depth of professional knowledge and a comprehensive understanding of existing solutions in the market. Studies show that high levels of product abandonment are due to several factors related to the quality of solution matching [18–20]. Providing personalised recommendations that accurately support the counselling process must be a procedure that takes into consideration the complex and rich, unique profile of every user, personality, abilities and needs, preferences, environment, tasks and goals. By using an AI-based matching algorithm, that calculates the matrix of every personal profile, the platform identifies and recommends the most suitable assistive technology solutions for the persons’ goals and needs. Thus, from the

wide range of tens of thousands of products available on the database, only the specific and most relevant products will appear in any search. The range of products that appear in the recommendations reflects the platform’s fundamental principles to open the market and to ensure comparisons of products from an empowered consumers’ position.

4. Holistic approach

When a person with disabilities or an older person seeks to drink, eat or communicate independently, and has a personal profile that expresses multiple challenges, capabilities and preferences, specific recommendations for cups and mugs, plates or mobile phones, will not be enough, even if they are a good fit themselves. Many potential challenges might be encountered, and so the types of assistive technology solutions that should be offered must represent a wide variety of commercial categories and product segments. An AI-based matching algorithm can provide this due to the computing capacity and ability to perform complex needs integration and interaction. Furthermore, this interaction between limitations and needs, can result in additional recommendations of assistive technologies and serve as an even more advanced part of the discovery, education and awareness of both assistive technology professionals and users.

5. The design of a digital tool should be a part of a wider service delivery model design

During product development Atvisor has encountered many questions regarding the service delivery model, from assessment to provision, that unless addressed, would serve as a barrier to the adoption of the platform. Creating high-level collaborations with governmental offices relevant to policy and regulation, while working with stakeholders on the ground to learn about their needs and the unique value propositions that are relevant to each has established the basis for the development of a unique service combining human resources and experience, together with digital tools. This development can eventually support the need for standardisation and scale in services, in line with global policy recommendations and systems thinking approach implemented in the field of assistive technology today [21,22].

6. The importance of building an online network infrastructure

The Atvisor platform offers an ongoing and dynamic interaction between assistive technology users, professionals and suppliers. For example, a user can reach out to their nominated professional online with questions or requests, and if necessary, can seek a professional from within the platform. The professional can share recommendations with the user and discuss them online, and the supplier can be approached for a purchase or a request regarding services. This network of interactions is highly relevant to the availability of assessment services in a period of increasing demand, to the discovery of assistive technology solutions and to the evolution of a new, much-needed awareness and proactive mindset that translates itself eventually to demand.

Even before the global pandemic, we recognised that the scale of unmet need for assistive technologies demanded innovation and change. Demographics and the aspirations of people with a disability, within a rights-based approach, meant that the traditional model of delivery could not fulfil demand. The impact of the pandemic has brought into sharp relief the issue that limited professional resources, including time, has delayed access to products and services leading to frustration and discrimination for people with a disability. The approach outlined and implemented

addresses those demands and offers an alternative approach based upon telerehabilitation with artificial intelligence.

Potential access to assistive technologies for the benefit of people with disabilities and older people could not look more promising, the diversity of products from social robots to smart homes to autonomous vehicles offers a world of options at our fingertips. These can help address many barriers. Our task now would be to ensure that assessment and decision-making equally respond to the possibilities, and that those who serve as gatekeepers to knowledge will have the proper “key” to make it accessible and personalised.

Today, where remote services become the principal channel for communication, this has never been more urgent.

Note

1. Pilots results and insight will be shared in upcoming articles.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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